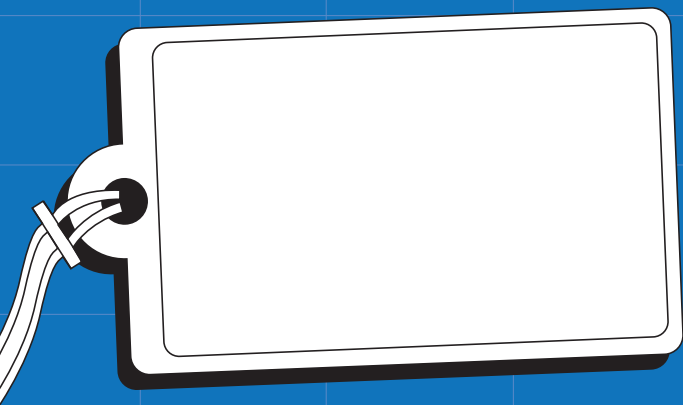




Factorising

FACTORISING

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FACTORISING

Factorising is the opposite of expanding. It is the process of putting expressions into brackets rather than expanding them out. In this section you will learn how different kinds of expressions are factorised in their own way.

Answer these questions, *before* working through the chapter.

I used to think:

What factors 20 and 12 have in common? What factors to $20ab$ and $12a$ have in common?

$(x - 2)(x + 2) = x^2 - 4$. Complete the following: $x^2 - 16 = (x - ?)(x + ?)$

How many brackets is a quadratic trinomial factorised into?

Answer these questions, *after* working through the chapter.

But now I think:

What factors 20 and 12 have in common? What factors to $20ab$ and $12a$ have in common?

$(x - 2)(x + 2) = x^2 - 4$. Complete the following: $x^2 - 16 = (x - ?)(x + ?)$

How many brackets is a quadratic polynomial factorised into?



What do I know now that I didn't know before?



Factorising

Factorising is the opposite of expanding. The goal of expanding is to remove brackets from expressions, like this example:

$$a(b + c) = ab + ac$$

The goal of factorising is to put expressions into brackets, like this example:

$$ab + ac = a(b + c)$$

Common Factors

Common factors are said to be 'factorised out' of expressions. In the expression $ab + ac$, the variable a is common in both terms. So a is a common factor of both terms. This means a can be 'factorised out' of the expression as above. Here are some examples:

Factorise these expressions:

a $5xy + 5y$
 $= 5y(x + 1)$

↑
 $5y$ is common in both terms

b $2x^2y - 4xy^2$
 $= 2xy(x - 2y)$

↑
 $2xy$ is common in both terms

In **a** above, both 5 and y are common factors on their own, but $5y$ is the **highest common factor (HCF)**. In **b**, $2xy$ is the highest common factor. To find the HCF, find the highest common numerical factor of the coefficients and then the **lowest** common power of each variable.

Factorise these expressions:

a $6a^5b^3 + 9a^3b^5$
 $= 3a^3b^3(2a^2 + 3b^2)$

$3a^3b^3$ is the HCF

- a is the HCF of the coefficients
- a^3 is the lowest power of a
- b^3 is the lowest power of b

b $12x^2y^5 - 4x^3y^3$
 $= 4x^2y^3(3y^2 - x)$

$4x^2y^3$ is the HCF

- 4 is the HCF of the coefficients
- x^2 is the lowest power of x
- y^3 is the lowest power of y

Sometimes there will be more than two terms. Then the HCF must be found for all the terms!

Factorise this expression:

$8a^3b^4 - 4a^3b^3 + 12a^2b^5$
 $= 4a^2b^3(2ab - a + 3b^2)$

↑

- 4 is the HCF of the coefficients
- a^2 is the lowest power of a
- b^3 is the lowest power of b

Brackets as Common Factors

Sometimes a bracket is a common factor. If this happens then the whole bracket is factorised out. Look at this example:

Factorise the following:

$$2x(x + 3) + 5y(x + 3)$$

(x + 3) is common in both terms

$$= (x + 3)(2x + 5y)$$

Factorising using Grouping and Brackets

The above example may not have been grouped into brackets, and may have been in expanded form

$$2x^2 + 6x + 5xy + 15y$$

Many expressions, like in the above line, could have 4 (or more) terms without a common factor across all terms.

Factorise the following:

a

$$2a^3 + 2ab + 4a^2b^2 + 4b^3$$

2a is common 4b² is common

$$= 2a(a^2 + b) + 4b^2(a^2 + b)$$

(a² + b) is common bracket

$$= (a^2 + b)(2a + 4b^2)$$

b

$$6x^2 - 9xy - 4x + 6y$$

3x is common -2 is common

$$= 3x(2x - 3y) - 2(2x - 3y)$$

(2x - 3y) is common bracket

$$= (2x - 3y)(3x - 2)$$

Sometimes the expression needs to be reordered before it can be grouped to find a common bracket. Here is an example:

Factorise the following into two brackets:

$$2a^2 + 3a - 2ab - 3b^2$$

Step 1: Reorder

$$= 2a^2 - 2ab + 3a - 3b^2$$

Step 2: Find common factors

$$= 2a(a - b^2) + 3(a - b^2)$$

Step 3: Factorise out the common bracket

$$= (a - b^2)(2a + 3)$$

Factorising by expanding the brackets to see if you find the original expression.

Find the HCF of these terms

a $6x^2$ and $2x$

b $8x^5$ and $6x^2y$

c $12a^2b^3$ and $8a^4b^2$

d $-4d^4e^3f^2$, $10d^2e^5f^3$ and $16d^5e^4f$

Factorise the HCF out of the following

a $6x + 2y$

b $2a^2 - 4a$

c $8pq^2 - 10q$

d $-3p^2q^4 + 12p^4q^2$

e $10x^2y^4 - 40x^4y^2$

f $8a^2b^5c^3 + 24a^5b^2c^6$

g $6p^2q^3 + 3p^3 - 15pq^4$

h $7mn^2 - 14m^5n^5 + 21m^3n^2$

Factorise the common bracket out of the following

a $3x^2(2x - 1) + 5y(2x - 1)$

a $4x(x^2y^3 - y^2) + 2(x^2y^3 - y^2)$

Factorise these expressions into two brackets using grouping

a $xy + y + 2x + 2$

b $ab^2 - 2ab + 4b - 8$

c $2x^2 + 3x + 4xy + 6y$

d $m^3n + m + 3m^2n + 3$

e $8xy^2 + 4x^2 - 4y^3 - 2xy$

f $2p^3 + q^2 + 3p^2q + 2pq$

g $4mn^2 + 3mn^4 - 4m - 3mn^2$

h $3y + xy^4 + y^2 + xy^3$

g $18p^3q + 12pq^2 + 6p^4q + 4p^2q^2$

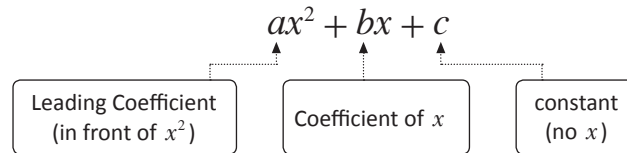
h $-2ab^2 + 4a^2b - 4a^2b^5 + 8a^3b^4$

Expanding brackets produces what is called a 'quadratic trinomial'. Look at this example

$$(x + 2)(x + 3) = x^2 + 5x + 6$$

The expression on the right is a quadratic trinomial because it has three terms and the highest power is 2.

In general, a quadratic trinomial looks like this



Remember, factorising is the opposite of expanding. So factorising a quadratic trinomial means changing it back to brackets. Factorising these trinomials is done differently depending on the value of a (the leading coefficient).

Factorising quadratic trinomials if $a = 1$

If $a = 1$ then the quadratic trinomial looks like this

$$x^2 + bx + c$$

To factorise these expressions, two numbers are needed such that:

1. Their product is c (the constant)
2. Their sum is b (the coefficient of x) and

These two numbers are written in the brackets. Here are some examples:

Factorise these quadratic trinomials into two brackets

a $x^2 + 5x + 6$

Step 1: Find two numbers with a product of 6 (the constant) and a sum of 5 (the coefficient of x).

$$\underbrace{2 \times 3}_{\text{Product}} = 6 \quad \text{and} \quad \underbrace{2 + 3}_{\text{Sum}} = 5$$

So the numbers are +2 and +3.

Step 2: Factorise by writing these two numbers in brackets with x .

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

b $x^2 - 5x - 6$

Step 1: Find two numbers with a product of -6 (the constant) and a sum of -5 (the coefficient of x).

$$\underbrace{1 \times (-6)}_{\text{Product}} = -6 \quad \text{and} \quad \underbrace{1 + (-6)}_{\text{Sum}} = -5$$

So the numbers are +1 and -6 .

Step 2: Factorise by writing these two numbers in brackets with x .

$$x^2 - 5x - 6 = (x + 1)(x - 6)$$

The order of these numbers is not important

Find values for m and n if they have these sums and products:

a $m + n = 3$ and $mn = 2$

b $m + n = 6$ and $mn = 4$

c $m + n = 0$ and $mn = -4$

d $m + n = -3$ and $mn = 2$

e $m + n = 3$ and $mn = -10$

f $m + n = -3$ and $mn = -18$

Answer these questions about the trinomial $x^2 + 9x + 20$

a Write down the value of the leading coefficient (a), the coefficient of x (b) and the constant (c)

b Find two numbers whose product is (c) constant and whose sum is (b)

c Factorise the trinomial into two brackets

Answer these questions about the trinomial $x^2 - 4x + 21$

a Write down the value of the leading coefficient (a), the coefficient of x (b) and the constant (c)

b Find two numbers whose product is (c) constant and whose sum is (b)

c Factorise the trinomial into two brackets

Factorise these trinomials into 2 brackets

a $x^2 + 4x + 3$

b $x^2 + x - 6$

c $x^2 - 6x + 5$

d $x^2 + 2x - 24$

e $x^2 - 6x + 16$

f $x^2 - 14x + 40$

g $x^2 - 5x + 14$

h $x^2 - 12x + 32$

Factorising quadratic trinomials if $a \neq 1$

Factorising $ax^2 + bx + c$ when a is not 1 is slightly more complicated. This will be shown with an example.

Factorise $12x^2 + 2x - 4$ into two brackets

Step 1: Find the value of $a \times c$

$$a \times c = 12 \times -4 = -48$$

Step 2: Find the two number which have a product of $a \times c$ (-48) and a sum of b (2)

$$8 \times (-6) = -48 \text{ and } 8 + (-6) = 2$$

So the two numbers are 8 and -6

Step 3: Rewrite the original b (coefficient of x) in terms of the two numbers in *Step 2*

$$\begin{aligned} 12x^2 + 2x - 4 \\ = 12x^2 + 8x - 6x - 4 \end{aligned}$$

Step 4: Using grouping to factorise

$$\begin{aligned} &= 4x(3x + 2) - 2(3x + 2) \\ &= (3x + 2)(4x - 2) \end{aligned}$$

Here is another example:

Factorise $12y^2 + 10y - 1$ into two brackets

Step 1: Find the value of $a \times c$

$$a \times c = 24 \times -1 = -24$$

Step 2: Find the two number which have a product of $a \times c$ (-24) and a sum of b (-10)

$$2 \times (-12) = -24 \text{ and } 2 + (-12) = -10$$

So the two numbers are 2 and -12

Step 3: Rewrite the original b (coefficient of y) in terms of the two numbers in *Step 2*

$$\begin{aligned} 12y^2 - 10y - 1 \\ = 12y^2 + 2y - 12y - 1 \end{aligned}$$

Step 4: Using grouping to factorise

$$\begin{aligned} &= 12y^2 - 12y + 2y - 1 \\ &= 12y(2y - 1) + (2y - 1) \\ &= (2y - 1)(12y + 1) \end{aligned}$$

Look at $16x^2 + 12x + 2$

- a Identify a , b and c
- b Find the value of $a \times c$
- c Find two numbers which have a product equal to $a \times c$ and a sum equal to b
- d Factorise $16x^2 + 12x + 2$ into two brackets.

Look at $6x^2 - x - 2$

- a Identify a , b and c
- b Find the value of $a \times c$
- c Find two numbers which have a product equal to $a \times c$ and a sum equal to b
- d Factorise $6x^2 - x - 2$ into two brackets.

Factorise these quadratic trinomials

a $2x^2 + 5x + 2$

b $12m^2 + 5m - 3$

c $14p^2 + 23p + 3$

d $5t^2 + 37t + 14$

e $8k^2 - 22k + 5$

f $6d^2 - 19x + 15$

Difference of squares

Remember when expanding brackets that differ in sign only $(x + y)(x - y) = x^2 - y^2$.

This means that if a quadratic expression is in the form $x^2 - y^2$ then it can be factorised easily into $(x + y)(x - y)$.

Here are some examples.

Factorise the following into two brackets using the difference of squares

a $x^2 - 16$

$$= x^2 - 4^2$$

$$= (x + 4)(x - 4)$$

b $4m^2 - 25n^2$

$$= (2y)^2 - (5n)^2$$

$$= (2y - 5n)(2y + 5n)$$

Don't get confused: $x^2 + y^2$ can't be factorised into two brackets. Only a difference of squares can be factorised this way.

Perfect Squares

Remember, for perfect squares $(x + y)^2 = (x + y)(x + y) = x^2 + 2xy + y^2$.

This means that if a quadratic expression is in the form $x^2 + 2xy + y^2$ then it can easily be factorised into $(x + y)^2$.

Here are some examples.

Factorise the following

a $m^2 + 6m + 9$

$$= m^2 + 2(m)(3) + 3^2$$

$$= (m + 3)^2$$

b $x^2 - 4x + 4$

$$= x^2 + 2(x)(-2) + (-2)^2$$

$$= (x - 2)^2$$

c $9c^2 + 12cd + 4d^2$

$$= (3c)^2 + 2(3c)(2d) + (2d)^2$$

$$= (3c + 2d)^2$$

d $9p^2 - 6pq + q^2$

$$= (3p)^2 + 2(3p)(-q) + (-q)^2$$

$$= (3p - q)^2$$

Whenever you factorise anything, you can always check by expanding the brackets to see if you get the original expression.

Factorise these expressions

c $x^2 + 14x + 49$

b $y^2 - 9$

c $4y^2 + 8y + 4$

d $16q^2 - 36p^2$

e $81p^2 - 121q^2$

f $64m^2 + 32mn + 4n^2$

g $49t^2 - 56tu + 16u^2$

h $9x^2y^2 + 12pqxy + 4p^2q^2$

Using more than one method to factorise

The methods learned up until now need to be combined sometimes to factorise certain expressions.

Factorise $4y^2 - 12y - 16$

Method 1

$$4y^2 - 12y - 16$$

Step 1: Factor out a common factor

$$= 4(y^2 - 3y - 4)$$

Step 2: Factorise the quadratic trinomial

$$= 4(y - 4)(y + 1)$$

Method 2

$$4y^2 - 12y - 16$$

Step 1: Factorise the trinomial

$$= 4y^2 - 16y + 4y - 16$$

$$= 4y(y - 4) + 4(y - 4)$$

$$= (y - 4)(4y + 4)$$

Step 2: Factorise the common factor out from the second bracket

$$= 4(y - 4)(y + 1)$$

$$\begin{aligned} -16 \times 4 &= -64 \quad (a \times c) \\ -16 + 4 &= -12 \quad (b) \end{aligned}$$

Sometimes the 'difference of squares' method needs to be used more than once, or combined with factoring out a common factor. Here are some examples.

Factorise the following

a $m^4 - n^4$

$$= (m^2)^2 - (n^2)^2$$

$$= (m^2 + n^2)(m^2 - n^2)$$

$$= (m^2 + n^2)(m + n)(m - n)$$

Remember: only a difference of squares can be factorised this way

b $3x^3 - 108x$

$$= 3x(x^2 - 36)$$

Common factor

$$= 3x(x + 6)(x - 6)$$

Difference of squares

If there are four terms or more, then grouping may have to be used with other methods. Here is an example.

Factorise $p^2 + 4p - q^2 + 4q$

$$p^2 + 4p - q^2 + 4q$$

$$= p^2 - q^2 + 4p + 4q$$

Difference of squares

$$= (p + q)(p - q) + 4(p + q)$$

$$= (p + q)[(p - q) + 4]$$

Common factor

$$= (p + q)(p - q + 4)$$

Factorising trinomials with more than one variable.

What happens if a quadratic trinomial like $x^2 - xy - 6y^2$ or $2x^2 + 7xy - 15y^2$ needs to be factorised? If the first and last terms are both squared, then the same process for factorising trinomials is followed.

Factorise $x^2 - xy - 6y^2$

Step 1: Find two numbers with a product of $-6y^2$ (coefficient of end term) and a sum of $-1y$ (coefficient of middle term)

$$\underbrace{-3y \times 2y = 6y^2}_{\text{Product}} \quad \text{and} \quad \underbrace{-3y + 2y = -1y}_{\text{Sum}}$$

So the numbers are $-3y$ and $+2y$.

Step 2: Factorise by writing these two numbers in brackets with x and y .

$$x^2 - xy - 6y^2 = (x - 3y)(x + 2y)$$

Factorise $2x^2 + 7xy - 15y^2$

Step 1: Find the value of the first coefficient and the last coefficient

$$2 \times -15 = -30$$

Step 2: Find the two numbers with a product of the above number and a sum of the middle coefficient

$$10 \times (-3) = -30 \quad \text{and} \quad 10 + (-3) = 7$$

So the two numbers are 10 and -3

Step 3: Rewrite the original middle coefficient in terms of the two numbers in **Step 2**

$$\begin{aligned} 2x^2 + 10xy - 3xy - 15y^2 \\ = 2x^2 - 3xy + 10xy - 15y^2 \end{aligned}$$

Step 4: Using grouping to factorise

$$\begin{aligned} &= x(2x - 3y) + 5y(2x - 3y) \\ &= (2x - 3y)(x + 5y) \end{aligned}$$

Sometimes, of course, this method of factorising will be combined with another.

Factorise $3p^3q + p^2q^2 - 4pq^3$

$$3p^3q + p^2q^2 - 4pq^3 = pq(3p^2 + pq - 4q^2)$$

$$= pq(3p^2 - 3pq + 4pq - 4q^2)$$

$$= pq[3p(p - q) + 4q(p - q)]$$

$$= pq(p - q)(3p + 4q)$$

$$\begin{aligned} -3 \times 4 &= -12 \quad (a \times c) \\ -3 + 4 &= 1 \quad (b) \end{aligned}$$

Factorise the following

a $10m^2 + 5m - 5$

b $-x^2 + 6x + 40$

c $x^8 - y^8$

d $8x^2 - 2xy - y^2$

e $28t^3 - 6t^2 - 10t$

f $y^2 + 6y - 2x^2 - 6x$

g $10x^3 - 1000x$

h $a^5 - ab^4$

i $-4x^2 - 2xy + 12y^2$

j $(y - 3)^2 - (y - 5)^2$

k $m^2 - 4n - 4n^2 - 2m$

l $-12t^2 + 27u^2 - 12t - 18u$

Algebraic Fractions

Numerical fractions can be simplified by finding common factors in the numerator and denominator. For example

$$\frac{12}{18} = \frac{2 \times 6}{3 \times 6} = \frac{2}{3}$$

Algebraic fractions have algebraic expressions in the numerator and the denominator. They can also be simplified by cancelling common factors. To do this, both the numerator and denominator need to be factorised.

Here are some examples:

Simplify these algebraic fractions

a $\frac{2x^2 - 8}{2x + 4}$

$$= \frac{2(x^2 - 4)}{2(x + 2)}$$

Common factor

$$= \frac{2(x + 2)(x - 2)}{2(x + 2)}$$

Difference of squares

$$= \frac{\cancel{2(x + 2)}(x - 2)}{\cancel{2(x + 2)}}$$

Cancel common brackets

$$= x - 2$$

b $\frac{y^2 - 2y - 15}{3y^2(y - 5) + 2(y - 5)}$

$$= \frac{(y + 3)(y - 5)}{3y^2(y - 5) + 2(y - 5)}$$

$$= \frac{(y + 3)(y - 5)}{(y - 5)(3y^2 + 2)}$$

$$= \frac{(y + 3)\cancel{(y - 5)}}{\cancel{(y - 5)}(3y^2 + 2)}$$

$$= \frac{y + 3}{3y^2 + 2}$$

To simplify any algebraic fraction, factorise the numerator and denominator and remove the common brackets and factors.

Simplify these algebraic fractions as much as possible

a $\frac{2x^3 - 3x^2 - 8x + 12}{2x^2 + x - 6}$

$$= \frac{x^2(2x - 3) - 4(2x - 3)}{(2x - 3)(x + 2)}$$

$$= \frac{(2x - 3)(x^2 - 4)}{(2x - 3)(x + 2)}$$

$$= \frac{\cancel{(2x - 3)}\cancel{(x + 2)}(x - 2)}{\cancel{(2x - 3)}\cancel{(x + 2)}}$$

$$= x - 2$$

b $\frac{(m - 2n)^2 - (2m + 3n)^2}{m^2 + 10mn + 25n^2}$

$$= \frac{[(m - 2n) - (2m + 3n)][(m - 2n) + (2m + 3n)]}{(m + 5n)^2}$$

$$= \frac{(-m - 5n)(3m + n)}{(m + 5n)(m + 5n)}$$

$$= \frac{\cancel{-(m + 5n)}(3m + n)}{\cancel{(m + 5n)}(m + 5n)}$$

$$= \frac{-(3m + n)}{m + 5n}$$

$$= \frac{-3m - n}{m + 5n}$$

Simplify these fractions as much as possible

a $\frac{2m^2 + 14m + 24}{m + 3}$

b $\frac{9x^2 - 4y^2}{9x^2 + 10xy + 4y^2}$

c $\frac{4t^3 - t^2 + 4t - 1}{4t^2 - 1}$

d $\frac{b^2 - c^2 + 4b + 4c}{b^2 - c^2}$

Multiplying and dividing fractions

Algebraic fractions are multiplied and divided the same way as numerical fractions. Before multiplying or dividing, factorise each fraction as much as possible. Then brackets can be cancelled to make calculations easier.

Here is an example multiplying two algebraic fractions

Multiply these two algebraic fractions

$$\frac{4a^2 - b^2}{b^2 + b - 20} \times \frac{b^2 - b - 12}{2a^2 - 5ab - 3b^2}$$

factorise the numerator and denominator in both fractions

$$= \frac{(2a + b)(2a - b)}{(b - 4)(b + 5)} \times \frac{(b + 3)(b - 4)}{(2a + b)(a - 3b)}$$

Cancel the common brackets

$$= \frac{\cancel{(2a + b)}(2a - b)}{\cancel{(b - 4)}(b + 5)} \times \frac{(b + 3)\cancel{(b - 4)}}{\cancel{(2a + b)}(a - 3b)}$$

$$= \frac{(2a - b)(b + 3)}{(b + 5)(a - 3b)}$$

Here is an example dividing two algebraic fractions

Find the following quotient

$$\frac{8x^2 - 6x - 5}{2x^2 + 7x + 3} \div \frac{4x - 5}{x^2 + 6x + 9}$$

Factorise the numerator and denominator in both fractions

$$= \frac{(2x + 1)(4x - 5)}{(x + 3)(2x + 1)} \div \frac{4x - 5}{(x + 3)^2}$$

Flip the second fraction to create a product of fractions

$$= \frac{(2x + 1)(4x - 5)}{(x + 3)(2x + 1)} \times \frac{(x + 3)^2}{4x - 5}$$

Cancel the common brackets

$$= \frac{(2x + 1)(4x - 5)}{(x + 3)(2x + 1)} \times \frac{(x + 3)^2}{4x - 5}$$

$$= x + 3$$

Adding and subtracting fractions

When adding and subtracting algebraic fractions, a common denominator must be found

Simplify the following

$$\begin{aligned} \text{a } \frac{4}{x} - \frac{3}{2x} \\ &= \frac{8}{2x} - \frac{3}{2x} \\ &= \frac{8-3}{2x} \\ &= \frac{5}{2x} \end{aligned}$$

$$\begin{aligned} \text{b } \frac{10}{y} + \frac{3y+1}{y^2} \\ &= \frac{10y}{y^2} + \frac{3y+1}{y^2} \\ &= \frac{10y+3y+1}{y^2} \\ &= \frac{13y+1}{y^2} \end{aligned}$$

Factorise the denominators, if possible, to make finding a common denominator easier.

Simplify the following

$$\begin{aligned} &\frac{2p^3 - 5p^2}{2p^2 - 3p - 10} + \frac{3}{p+2} \\ &= \frac{2p^3 - 5p^2}{(p+2)(2p-5)} + \frac{3}{p+2} \quad \leftarrow \text{Factorise denominator} \\ &= \frac{2p^3 - 5p^2}{(p+2)(2p-5)} + \frac{3(2p-5)}{(p+2)(2p-5)} \quad \leftarrow \text{Common denominator} \\ &= \frac{2p^3 - 5p^2 + 3(2p-5)}{(p+2)(2p-5)} \\ &= \frac{2p^3 - 5p^2 + 6p - 15}{(p+2)(2p-5)} \\ &= \frac{p^2(2p-5) + 3(2p-5)}{(p+2)(2p-5)} \quad \leftarrow \text{Grouping in pairs} \\ &= \frac{(2p-5)(p^2+3)}{(p+2)(2p-5)} \\ &= \frac{\cancel{(2p-5)}(p^2+3)}{(p+2)\cancel{(2p-5)}} \quad \leftarrow \text{Cancel common brackets} \\ &= \frac{p^2+3}{p+2} \end{aligned}$$

Simplify the following products and quotients

a $\frac{2y - 2}{3x - 3y} \times \frac{3x^2 - 3xy}{2y^2 - 2y}$

b $\frac{2m^2 + 13m - 7}{m^2 - 49} \times \frac{2m^2 - 14m}{2m - 1}$

c $\frac{p + 5}{p + 2} \div \frac{p^2 + 10p + 25}{p^2 - 4p + 4}$

d $\frac{4 - x^2}{6 - 5x + 4x^2} \div \frac{2 - x}{6 - 11x + 4x^2}$

e $\frac{4a^2 - b^2}{6a^2 - 11ab + 4b^2} \times \frac{3a - 4b}{a - b}$

e $\frac{4x - 3y}{x + 2y} \div \frac{8x^2 + 6xy + 9y^2}{4x^2 - 9y^2}$

Simplify these expressions

e $\frac{3}{2a} - \frac{5}{3a}$

e $\frac{2x^2 - 3}{6x} - \frac{x + 1}{4}$

c $\frac{4x+2}{x+1} + \frac{x-4}{2x+2}$

d $\frac{2y}{y^2-1} + \frac{3y}{y+1}$

e $\frac{3}{x+2} - \frac{3}{2x+1}$

f $\frac{2}{(y+3)(y+4)} - \frac{1}{(y+3)(y-4)}$

g $\frac{2p}{3p^2 + 5p + 2} - \frac{p}{p + 1}$

h $\frac{m^2 - 100}{m^2 + 13m + 30} + \frac{m - 3}{2m^2 - 18}$

i $\frac{1}{x+1} + \frac{1}{x+2} + \frac{1}{x-1}$

j $\frac{4a-2b}{3a-5b} \div \frac{4a+6b}{6a^2-ab-15b^2}$

